

Formalization of Gödel’s Incompleteness Theorems in Basic Recursive Arithmetic (BRA)

Summary of an Agda development

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Abstract

We present a fully formalized proof of both Gödel’s First and Second Incompleteness Theorems in a tree-based variant of Guard’s Basic Recursive Arithmetic (BRA), executed in Agda 2.8 with `--safe --without-K --exact-split`. The development contains **zero postulates and zero holes**; every per-file solo typecheck completes in under one second (with one historical exception, the 16k-line dispatcher cascade for *thmT*).

The headline results are:

- *thm11* : *Deriv G* → *Deriv* ⊥ (Gödel I: provability of the diagonal sentence implies ⊥);
- *godelII* : *Deriv Con* → *Deriv* ⊥ (Gödel II: provability of consistency implies ⊥).

The development follows Guard’s 1963 lecture notes on recursive arithmetic exclusively. The intermediate construction is a single first-order computable function *constr5* : *Fun1* realizing Guard’s “*F(x)*” (page 17 of his notes); each step in the chain is given by an explicit equational derivation in BRA.

1 Introduction

The aim of the project is to give a feasible, ASCII-only, syntax-first Agda formalization of Gödel’s two incompleteness theorems, following Guard’s 1963 system of *Basic Recursive Arithmetic* [1]. The intent is to avoid historical baggage (Gödel numbering of Hilbert proofs, primitive recursive arithmetic with explicit numerals, “encoded” substitution as a Hilbert-system meta theorem) by working inside a small tree-based equational system from the outset:

- Terms are built from *O* (leaf), *var n*, *ap₁ f t*, *ap₂ g t₁ t₂*.
- Function symbols come in two arities: *Fun₁* and *Fun₂*, with primitive constants {*I*, *Fst*, *Snd*, *Comp*, *Comp₂*, *Z*} and {*Pair*, *Const*, *Lift*, *Post*, *Fan*, *IfLf*, *TreeEq*, *treeRec*}.
- Equations are objects *eqn ℓ r* between terms; formulas *Formula* extend equations with *not* and *imp*.
- Provability is an inductive type *Deriv F* with constructors for the algebraic axioms (*ax_{*}*), the equality axioms, the propositional axioms (*ax_K*, *ax_S*, *ax_{Neg}*, *ax_{ExFalso}*, *ax_{Contrapos}*), modus ponens (*mp*), substitution (*ruleInst*), and binary tree induction (*ruleIndBT*).

The natural numbers are represented as binary trees (*lf* for 0, *nd lf n* for the successor in unary). The encoding *reify* : *Tree* → *Term* maps *lf* ↦ *O* and *nd a b* ↦ *ap₂ Pair (reify a) (reify b)*.

2 The provability predicate $thmT$

The Gödel-style “ th ” provability predicate is realized in BRA by a single Fun_1 -functor:

$$thmT = Rec\ stepProtoWrapped : Fun_1.$$

Here $stepProtoWrapped$ is a Fun_2 assembled from a 40-deep linear $IfLf$ cascade (BRA/Thm/StepProto.agda and BRA/Thm/ThmT.agda, ~16k lines), one branch per primitive proof rule: ten equational axioms of Group I (axI, axFst, ..., axKT), eight Group II axioms (axIfLfL, axIfLfN, four axTreeEq*, axGoodstein), nine equality and propositional rules, and three rules of inference (mp , $ruleInst$, $ruleIndBT$).

The *key* property of $thmT$ is that on the encoded form of any proof tree it returns the encoded conclusion of that proof. This is the content of the *encoding-soundness* lemma $encode : (P : Formula) \rightarrow Deriv\ P \rightarrow \Sigma\ Tree\ (\lambda y \rightarrow Deriv\ (thmT\ (reify\ y) = reify\ (codeFormula\ P)))$, proved by induction on the $Deriv$ derivation in BRA/Thm/ThmT.agda.

2.1 Self-substitution closure

A second key property is that $thmT$ contains no free variable (it is a closed combinator):

$$thClosed : \forall x\ r, \quad substF_1\ x\ r\ thmT = thmT$$

and that the encoding of $thmT$ is var-free:

$$codeF1Th_noVar : \forall x\ repl, \quad codedSubst\ repl\ (code\ (var\ x))\ (codeF_1\ thmT) = codeF_1\ thmT.$$

Both are `refl` inside the abstract block where $thmT$ unfolds.

3 Gödel’s First Incompleteness Theorem (Theorem 11)

3.1 The diagonal sentence

Following Guard, define

$$\begin{aligned} F_{pre} &:= \neg(thmT\ (v_1) = sub\ (v_0)\ (v_0)), \\ j &:= codeFormula\ F_{pre} \quad (\text{a closed Tree}), \\ G &:= F_{pre}[v_0 := reify\ j] = \neg(thmT\ (v_1) = sub\ i\ i), \end{aligned}$$

where $i := reify\ j$ and sub is the BRA-internal coded substitution function (BRA/Sub.agda). The sentence G has v_1 free; informally, G asserts of itself that no proof tree v_1 proves the formula whose code is $j[v_0 := i]$.

3.2 Theorem 11

Pulling apart the abstract closure (BRA/Thm11Abstract.agda), the precise statement is:

Theorem 1 (thm11, Gödel I). $thm11 : Deriv\ G \rightarrow Deriv\ \perp$, where $\perp := atomic\ (eqn\ trueT\ falseT)$ with $trueT := O$ and $falseT := ap_2\ Pair\ O\ O$.

Sketch. From any $\pi : Deriv\ G$ extract its encoding (y_d, E_1) where $E_1 : Deriv\ (thmT\ (reify\ y_d) = reify\ (codeFormula\ G))$. A separate *diagonal lemma* **diagonal** (BRA/Thm11Diagonal.agda) yields $N : Deriv\ \neg(thmT\ (reify\ y_d) = reify\ (codeFormula\ G))$ from π , and $ax_{ExFalse} +$ two mp ’s collapse E_1 and N to $Deriv\ \perp$. $\square$

The diagonal lemma is constructed inside an abstract block in `BRA/Thm11Diagonal.agda` using `thClosed`, `codeF1Th_noVar`, and `cc` (the local instance of the meta-equation `codedSubst (code i) (code v0) j = codeFormula G`, which is provable by induction on F_{pre} via `codeCommutesFormula` in `BRA/CodeCommutes.agda`).

4 Gödel’s Second Incompleteness Theorem (Theorem 14)

The second theorem is the more substantial construction. It says that the consistency formula *Con* entails $\perp$:

Theorem 2 (`godelIII`, Gödel II). *godelIII* : $\text{Deriv } \text{Con} \rightarrow \text{Deriv } \perp$, where $\text{Con} := \neg(\text{thmT } (v_0) = \text{reify } (\text{codeFormula } \perp))$ (consistency: “no proof, encoded as v_0 , derives $\perp$ ”).

The construction follows Guard’s pages 16–17 step by step.

4.1 Numerals: the asymmetry $f(\underline{x})$ vs. $\underline{f(x)}$

A crucial feature of the Theorem 12–14 chain, easy to miss in the formalisation, is the *asymmetric* role of the numeral function. In Guard’s notation [1] the underline $\underline{x}$ denotes the *numeral* of x : a closed BRA term whose value is x . In our Agda development this is the primitive-recursive function

$$\text{num} = \text{cor} : \text{Tree} \rightarrow \text{Tree},$$

which sends a tree x to the code of the BRA term that *names* x . We use *num* and *cor* interchangeably with Guard’s $\underline{x}$.

The asymmetry is this: for any Fun_1 -functor f and tree x ,

- $\text{ap}_1 f (\text{num } x)$, i.e. $f(\underline{x})$, is the code of a BRA term — specifically the term “ f applied to the numeral of x ”;
- $\text{num } (f x)$, i.e. $\underline{f(x)}$, is *generally not* the code of any such syntactic term — it is the code of the numeral naming the *value* $f x$.

The two trees disagree in general; they agree only after BRA proves them equal.

Theorem 12 is precisely the internalisation of that equation: for every $f : \text{Fun}_1$ it builds a $D_f : \text{Fun}_1$ such that

$$\text{Deriv } (\text{thmT } (D_f x) = \text{codeFTeq}_1 f x),$$

where the right-hand side $\text{codeFTeq}_1 f x$ is the encoded equation $\text{ap}_1 f (\text{cor } x) = \text{cor } (f x)$, i.e. the encoded form of $f(\underline{x}) = \underline{f(x)}$. In other words: BRA itself proves the bridge between the syntactic side (left-hand: a code-of-term) and the semantic side (right-hand: the numeral of the value). This is what licenses every subsequent step of Goedel II: it lets the internal *thmT* predicate move between the two encodings exactly where Guard’s pencil-and-paper argument tacitly does so.

A symmetric variant in which both sides are coded the same way (*code* $(f x)$ on both sides, say) collapses: the resulting trees do not align across the five steps of Theorem 14, and step 5 becomes underivable. The asymmetric form $f(\underline{x}) = \underline{f(x)}$ is therefore not a presentational choice but a load-bearing feature of the proof.

4.2 Theorem 12 — the encoded “*th f*” clause

For every Fun_1 -functor f , Theorem 12 produces a Fun_1 D_f such that under the canonical encoding, $thmT$ evaluated at $D_f x$ is provably equal to the encoded form of “ $f(num\ x) = num\ (f\ x)$ ”. Formally:

Theorem 3 ($thm12_F_1$, parametric Theorem 12 for Fun_1). $thm12_F_1 : (f : Fun_1) \rightarrow Thm12_F_1_Result\ f$ where the record $Thm12_F_1_Result\ f$ packages a $D_f : Fun_1$ together with $proof : (x : Term) \rightarrow Deriv\ (thmT\ (D_f\ x) = codeFTeq_1\ f\ x)$. The conclusion $codeFTeq_1\ f\ x$ is the literal Pair-shape encoded equation $ap_1\ f\ (cor\ x) = cor\ (f\ x)$.

(Analogous statement for Fun_2 .) This is a structural recursion on f : there is one clause per Fun_1 / Fun_2 constructor (15 total), each in its own `BRA/Thm12/Parts/*.agda` file. The composite cases (*Comp*, *Comp₂*, *Lift*, *Post*, *Fan*, *treeRec*) dispatch to per-case modules; the recursive *TreeEq* case uses `ruleIndBT2` (2-tree-pair induction with diagonal cross-IHs) to establish closure. The whole theorem is in `BRA/Thm12.agda`, module `FromBridges`, and contains zero postulates / zero holes.

4.3 Theorem 13 — under hypothesis “ $f\ x = y$ ”

Building on Theorem 12, one produces under the hypothesis $f\ x = y$ the encoded form of “ $f(num\ x) = num\ y$ ”:

Lemma 4 ($thm13_F_1$). For every $f : Fun_1$ and every Theorem 12 result r_{12} for f , $thm13_F_1\ f\ r_{12}\ x\ y : Deriv\ (f\ x = y) \rightarrow Deriv\ (thmT\ (D_f\ x) = codeFTeq_{1,Hyp}\ f\ x\ y)$ where the conclusion replaces the “ $cor\ (f\ x)$ ” slot of $codeFTeq_1$ by $cor\ y$. One line: `ruleTrans` on $r_{12}.proof\ x$ with `congR Pair` on `cong1 cor` of the hypothesis.

4.4 Theorem 14 — the consistency-to-bot construction

This is the heart of Gödel II. The contract abstracted in `BRA/Thm14Abstract.agda` asks for:

1. a Fun_1 -functor $constr5$ such that $thmT\ (constr5\ x)$ syntactically encodes $\perp$;
2. a parametric internal-implication

$step5 : (x : Term) \rightarrow Deriv\ ((thmT\ x = reify\ (codeFormula\ G))\ imp\ (thmT\ (constr5\ x) = reify\ (codeFormula\ G)))$

The closure $closureToG$ then applies $ax_{Contrapos}$ to $step5$ instantiated at v_1 , $ruleInsts\ Con$ at $constr5\ v_1$, modus-ponenses to obtain $\neg(thmT\ v_1 = i)$, transports through $subIteqG\ (sub\ i\ i = cor\ G)$ and G_{unfold} (propositional normal form of G), arriving at $Deriv\ G$. Composing with $thm11$ gives $godelII : Deriv\ Con \rightarrow Deriv\ \perp$.

4.4.1 Construction of $constr5$

Following Guard’s page 17, the construction relies on two pre-existing tautologies, encoded:

- $t := (v_0 = v_2)\ imp\ ((v_1 = v_2)\ imp\ (v_0 = v_1))$ (transitivity);
- $t' := (v_0 = v_1)\ imp\ ((\neg(v_0 = v_1))\ imp\ \perp)$ (ex falso).

Both are derivable in BRA (BRA/Thm14Numerals.agda), and via *encode* they yield closed Trees *tterm*, *t'term* such that *thmT* (*reify tterm*) = *reify (codeFormula t)* (and analogously for *t'*).

The chain is:

$$\begin{aligned}
f_part(x) &:= ruleInst\ v_0\ (th\ \hat{x})\ (ruleInst\ v_1\ (sub\ i\ i)\ (ruleInst\ v_2\ cor\ G\ tterm)), \\
g_part(x) &:= mp(mp(f_part(x), D_{thmT}\ x), D_{sub}\ i\ i), \\
K_part(x) &:= ruleInst\ v_1\ (cor\ x)\ x, \\
h_part_skel(x) &:= ruleInst\ v_0\ (th\ \hat{x})\ (ruleInst\ v_1\ (sub\ i\ i)\ t'term), \\
constr5_final(x) &:= mp(mp(h_part_skel(x), g_part(x)), K_part(x)).
\end{aligned}$$

The substituents are the encoded mixed-form Pairs of Guard: $\widehat{th\ \hat{x}} := ap_2\ Pair\ (reify\ tagAp_1)\ (ap_2\ Pair\ (reify\ (code\ sub\ i\ i)))$, and $cor\ G := reify\ (code\ (reify\ (codeFormula\ G)))$.

4.4.2 The five steps and their lemmas

Following Guard's enumeration (guard15.pdf pages 16–17):

Step 1 — *step1_l* (Th14Constr5.agda). By Theorem 13 applied to *thmT* at $(x, cor\ G)$ under hypothesis $thmT\ x = cor\ G$, get *Deriv* (*PrAtX* *x imp thmT* (*D_{thmT}* *x*) = *codeFTeq_{1,Hyp}* *thmT* *x cor G*).

Step 2 — *step2_l* (Th14Step2.agda). By Theorem 13 applied to *sub* at $(i, i, cor\ G)$ under closed hypothesis *subIleqcG*.

Step 3 — *step3_l* (= *g_part_l*) (Th14Step3.agda, Th14Step3Canon.agda). Three sb-applications on *t* via *thmTSubLemma*, canonicalized via *subT_codeFormula_imp/atomic* and leaf preservation lemmas, then two *mp* dispatches at *g_part_inner* and *g_part*. Output: *Deriv* (*PrAtX* *x imp thmT* (*g_part* *x*) = $th\ \widehat{\hat{x} = sub\ i\ i}$).

Step 4 — *K_part_l* (Th14Step4Final.agda). One sb-application at *v₁* on the diagonal sentence *G_{norm}* via *thmTSubLemma* + the *PrAtX* *x* hypothesis to swap $thmT\ x \mapsto cor\ G$, canonicalized via *subT_codeFormula_neg/atomic* and leaf preservation. Output: *Deriv* (*PrAtX* *x imp thmT* (*K_part* *x*) = $th\ \widehat{\hat{x} \neq sub\ i\ i}$).

Step 5 — *step5_l* (Th14Step5.agda). Two outer *mp* dispatches at *constr5_inner*, *final* and *constr5_final*, with *step3_l* feeding the inner antecedent and *K_part_l* standing in syntactically as the second-*mp* witness. The dispatcher's syntactic projection of the consequent gives the encoded $\perp$. Output: *Deriv* (*PrAtX* *x imp thmT* (*constr5_final* *x*) = *reify (codeFormula $\perp$)*).

Auxiliary — *h_part_skel_l* (Th14Step5HPart.agda). Two sb-applications on *t'*, mirror of step 3 (one fewer layer because *t'* has only two free variables). Output: *thmT* (*h_part_skel* *x*) = *ex falso* form. Used as *step5_l*'s inner *d₁*.

The headline composition (BRA/GoedelIIIFull.agda) is then:

```

r12_th  : Thm12_F1_Result thmT
r12_th  = FromBridges.thm12_F1 thmT

r12_sub : Thm12_F2_Result sub
r12_sub = FromBridges.thm12_F2 sub

```

```

module Step5 = BRA.Th14Step5.Th14Step5 r12_th r12_sub
module Final = BRA.GoedelIII.Compose Step5.constr5_final Step5.step5_l

```

```

godelIII : Deriv Con -> Deriv bot
godelIII = Final.godelIII

```

4.5 Modal-logic perspective: Theorems 12–14 as $\Box$ -functoriality, necessitation, and the diagonal

The asymmetry of Section 4.1 and the structure of Theorems 12–14 admit a clean reading in intuitionistic modal $S4$. Following Davies and Pfenning [3], separate *value variables* (in Γ) from *code variables* (in Δ) and write $\Box A$ for “code of type A ”, with introduction **box** $E : \Box A$ when $\Delta; \cdot \vdash E : A$ (only modal-variable dependencies allowed) and elimination **let box** $u = E_1$ **in** E_2 . This two-context discipline is precisely the truth/validity distinction.

Read in this light:

The *cor/code* asymmetry: $f(\underline{x})$ vs $f(x)$. In the modal calculus $\Box A$ and A are different types, and a value-level function $f : \overline{A} \rightarrow B$ cannot be applied to a code: **box** $(f\ x) : \Box B$ is well-typed, but f (**box** x) is a type error. In BRA the corresponding pair of expressions are $\underline{f(x)} = \text{cor}(f\ x)$ (always a code by construction) and $f(\underline{x}) = f(\text{cor } x)$ (just some tree, computed by running f on the numeral of x ; not of the form $\text{cor } z$ for any natural z). Both inhabit the uniform type *Tree* in BRA, so no type error fires — but the modal status differs, and that difference is exactly the content of Theorem 12. The slogan “ $\underline{x}$ is the code of x , but $f(\underline{x})$ need not be the code of anything” is the BRA shadow of Davies–Pfenning’s typing distinction.

Theorem 12 is the functoriality of $\Box$ over $\text{Fun}_1 / \text{Fun}_2$. At the BRA level a Fun_1 -functor f is a primitive recursive map $A \rightarrow B$ (here $A = B = \text{Tree}$). Theorem 12 produces a $D_f : \text{Fun}_1$ realising the lifted map $\Box A \rightarrow \Box B$, witnessed by an internal derivation of the equation

$$\text{thmT } (D_f\ x) = \widehat{f(\underline{x})} = \underline{f(x)}.$$

The two sides of the encoded equation are the two paths $x \mapsto f(\text{cor } x)$ and $x \mapsto \text{cor}(f\ x)$ in what would be the unit-naturality square of $\text{cor} : \text{id} \Rightarrow \Box$. A subtlety: Theorem 12 doesn’t *assume* a pre-existing functor $\Box$ and prove naturality on top — it simultaneously *defines* the functor’s action on morphisms (the D_f) and proves the square commutes. Functoriality and naturality are produced in one move.

Theorem 13 is necessitation for equations. Given a value-level *Deriv* $(f\ x = y)$ it produces an internal derivation of $\text{thmT } (D_f\ x) = \widehat{f(\underline{x})} = \underline{y}$, replacing $f(x)$ by $\underline{y}$ inside the box. This is the Hilbert–Bernays clause D1 ($\vdash A \Rightarrow \vdash \Box A$) restricted to equational A , composed with Theorem 12’s naturality to rewrite the $\underline{f(x)}$ side of the encoded equation as $\underline{y}$. It is not the modal K axiom $\Box(A \rightarrow B) \rightarrow \Box A \rightarrow \Box B$; the K -flavoured rule is Theorem 12 itself (lifting the morphism f to act on boxed inputs), whereas Theorem 13 is the downstream necessitation step that closes equations under the functor.

Theorem 14, step 5, internalises the unprovability half of Gödel I at the diagonal. *step5_l* derives, parametrically in x , the encoded internal implication

$$\text{thmT } x = \text{cor } G \quad \text{imp} \quad \text{thmT } (\text{constr5 } x) = \text{cor } \perp.$$

Read externally: “if x codes a proof of G , then $\text{constr5 } x$ codes a proof of $\perp$ ”. The derivation uses two ingredients. First, the diagonal property of G at the encoded level ($G \leftrightarrow \neg \text{Pr}(\ulcorner G \urcorner)$): any proof of G commits the system to its own unprovability statement. Second, the Hilbert–Bernays clause D2 ($\Box A \rightarrow \Box \Box A$, “provable propositions are provably provable”) applied at $A := \text{Pr}(\ulcorner G \urcorner)$, which lets the contradiction be *internalised* rather than merely observed. The combination is the standard internal version of the Gödel I direction $\text{Con} \Rightarrow \neg \text{Pr}(\ulcorner G \urcorner)$, packaged as $\text{Pr}(\ulcorner G \urcorner) \rightarrow \text{Pr}(\ulcorner \perp \urcorner)$. This is *not* Löb’s theorem ($\Box(\Box A \rightarrow A) \rightarrow \Box A$); Löb is not used anywhere in the chain. constr5 is the explicit primitive recursive realiser of this internalised Gödel I clause at the diagonal G ; the chain f_part , g_part , K_part , h_part_skel assembled in Section 4.1 above is the syntactic decomposition of that realiser.

What this perspective does *not* do is shorten the BRA proof. Davies and Pfenning’s $\Box$ is a metatheoretic construct; their typing rules leave $\Box A$ abstract and give no algorithmic content for D_f , constr5 , or thmT . Gödel II proper requires the provability predicate to be a primitive recursive function of the object theory, definable inside BRA itself — the deep encoding is intrinsic to the theorem, not an artefact of the formalisation. The modal reading is therefore best understood as a *specification* that the BRA construction realises: theorem by theorem, BRA delivers the content that an $S4$ -style code modality would require, via an explicit primitive-recursive witness.

5 Methodology

5.1 Fast typecheck as a correctness oracle

Every per-file solo typecheck runs in well under one second (with the historical exception of BRA/Thm/ThmT.agda at $\sim 19\text{s}$; this is the 16k-line dispatcher cascade and was already in place before the Gödel II push). The session followed a strict rule: *slow typecheck = mathematical mismatch, stop and re-architect*. In practice this prevented two specific traps:

1. Trying to feed Agda an opaque normalisation it cannot perform (e.g. unfolding *codeFormula* G at a parametric substituent position). The fix is always: add a single targeted refl-witnessable lemma inside the right **abstract** block (where the opaque definition unfolds), and use it as a black box outside.
2. Forcing structural recursion through opaque trees (*codedSubst/codedSubstT* on *codeF₁ thmT*, etc.). Same fix: prove the equation once inside the abstract block, expose only the type signature outside.

5.2 Architectural choice for *constr5*

The original `Th14Constr5Final.constr5` wraps *h_part* with two *additional* encoded-mp layers (*h_part_thxLayer*, *h_part*). Under the *thmT* dispatcher these would consume both antecedents of *h_part_skel*’s ex-falso shape, collapsing the value of *thmT* (*h_part x*) to $\perp$ before it can serve as antecedent to *constr5_inner*. The fix is to use *h_part_skel* directly in *constr5_final* (pure sb-chain, no internal mp’s), matching Guard’s intent on page 17 and the comment in `Th14Constr5.agda`. The two outer *mp*’s in *constr5_final* then consume the two antecedents exactly as Guard’s chain prescribes.

5.3 The Carneiro-lifted chain: simulating the deduction theorem in a Hilbert system

BRA’s *Deriv* is a Hilbert-style system: implication is a connective, but there is no native introduction rule that, given a derivation of ψ depending on a hypothesis φ , returns *Deriv* ($\varphi \text{ imp } \psi$). The propositional axioms are Guard’s

$$axK = P \text{ imp } (Q \text{ imp } P), \quad axS = (P \text{ imp } (Q \text{ imp } R)) \text{ imp } ((P \text{ imp } Q) \text{ imp } (P \text{ imp } R)),$$

together with modus ponens. *axS* is Frege’s 1879 axiom.

In such a system the deduction theorem can be *simulated* (a technique attributed to Mario Carneiro in the metamath community): replace every theorem ψ_i in a derivation sequence by $\varphi \text{ imp } \psi_i$. If ψ_k follows from ψ_i and $\psi_j = \psi_i \text{ imp } \psi_k$ by modus ponens, then $\varphi \text{ imp } \psi_k$ follows from $\varphi \text{ imp } \psi_i$ and $\varphi \text{ imp } (\psi_i \text{ imp } \psi_k) = \varphi \text{ imp } \psi_j$ via *axS* and two applications of modus ponens. Frege’s axiom is precisely what licenses the lifting; the form was apparently understood in 1879 and lost in Russell’s and Hilbert’s later presentations.

The Theorem 14 chain uses this trick at the level of the running hypothesis

$$PrAtX\ x := (thmT\ x = cor\ G).$$

Each step lemma *step1_l*, ..., *step5_l*, *K_part_l*, *h_part_skel_l* is the *PrAtX x imp (...)*-lifted form of the underlying step equation; the suffix “_l” (*lifted*) marks this throughout the codebase. Modus ponens between two adjacent step lemmas is mediated by *axS*. The reusable combinators sit in *BRA/Thm/EvalHelpers.agda* (header “Lift helpers (deduction-theorem combinators)”); the lifted Theorem 13 for singular functors is in *BRA/Th14Constr5.agda* (“Carneiro-lifted Theorem 14 chain”); the per-step proofs live in *BRA/Th14Step3.agda*, *Th14Step4Final.agda*, *Th14Step5.agda*, and *Th14Step5HPart.agda*. Theorem 12’s *Rec / RecP* cases use the same idea at *dispatcher granularity* — the per-tag dispatcher is what is lifted, not each individual equation (*BRA/Th12RecP.agda*, *BRA/TreeEqReflParam.agda*, with helpers in *BRA/Thm/EvalHelpers.agda*).

Without this simulation the Theorem 14 development would not close: each step genuinely depends on *PrAtX x*, and a Hilbert-style *Deriv* offers no native means to discharge such a hypothesis. The Carneiro lifting is therefore not a stylistic choice but a structural component of the proof, on a par with the *cor/code* asymmetry of Section 4.1.

6 Statistics

6.1 File-by-file summary of the Gödel II push

File	LoC	Solo typecheck
BRA/Th14SubTLeaf.agda	374	0.18 s
BRA/Th14Step3Canon.agda	582	0.60 s
BRA/Th14Step3.agda	633	0.81 s
BRA/Th14Step4Final.agda	213	0.84 s
BRA/Th14Step5HPart.agda	394	0.84 s
BRA/Th14Step5.agda	305	0.76 s
BRA/GoedelIIIFull.agda	50	0.71 s

6.2 Earlier infrastructure (already in place)

- *BRA/Thm/ThmT.agda*: 16k LoC, 40-deep dispatcher cascade, 19 s.

- BRA/Thm12.agda: structural Theorem 12 with one Parts file per constructor, 0.65 s.
- BRA/Thm13.agda: ~ 80 LoC, 0.1 s.
- BRA/Thm11.agda, Thm11Abstract, Thm11Diagonal: ~ 300 LoC total.
- BRA/Thm14Abstract.agda: ~ 250 LoC, the *closureToG* machinery.
- BRA/Thm14Constr5Final.agda, Th14StepHPre, Th14HUnfolds, Th14GPartUnfolds: ~ 700 LoC of Pair-of-Comp2 unfold lemmas.

6.3 Constraints maintained throughout

- `--safe --without-K --exact-split` on every file.
- ASCII only. No Unicode.
- No postulates. No holes. No `with`-abstraction. No dot patterns.
- No new *Deriv* axioms beyond Guard’s primitive list.

7 Discussion

The two Gödel theorems in BRA, side by side:

- **Gödel I.** If the diagonal sentence G is provable, then $\perp$ is provable. Equivalently (under the meta-assumption that BRA is consistent): G is unprovable.
- **Gödel II.** If the consistency formula Con is provable, then $\perp$ is provable. Equivalently: BRA cannot prove its own consistency.

The intermediate functions $constr5_{\text{final}}$ and the parametric step lemmas $step1_l\text{--}step5_l$, K_part_l , $h_part_skel_l$ are *first-order* in the sense of recursive-arithmetic definability: each is a closed combinator built from primitive Fun_1 / Fun_2 symbols, and each *Deriv* derivation is given by an explicit equational proof in the BRA system. No external soundness theorem is invoked.

7.1 Comparison with prior formalisations

To our knowledge the only previous machine formalisation of Gödel’s *Second* Incompleteness Theorem is Paulson’s Isabelle/HOL development of hereditarily finite set theory [2]. Other formalisations (O’Connor in Coq, Han in Lean, Carneiro’s metamath work) target Gödel’s *First* theorem only.

Paulson’s development uses HF set theory and full Gödel numbering of Hilbert-style proofs, with encoded substitution treated as an external soundness theorem. The present BRA development by contrast follows Guard [1]: the “provability predicate” $thmT$ is itself a primitive-recursive combinator built from $Rec + Fan + Lift + IfLf$; encoded substitution is performed by an explicit primitive recursive function $sub : Fun_2$; and the entire chain (Gödel I and II) stays first-order, no Hilbert-proof Gödel numbering, no external soundness theorem.

References

- [1] J. R. Guard, *Lecture notes on recursive arithmetic*, Applied Mathematics Division, Argonne National Laboratory, 1963.
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- [3] R. Davies and F. Pfenning, *A modal analysis of staged computation*, Journal of the ACM, vol. 48 no. 3, May 2001, pp. 555–604.